

Changing Dimension in FFGFT

Geometric Reduction ($T^4 \rightarrow T^0$) and Representational Translation

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Abstract

The reduction of the fundamental 4-torus structure T^4 of FFGFT to lower dimensions is examined as a formal chain $T^4 \rightarrow T^3 \rightarrow T^2 \rightarrow T^1 \rightarrow T^0$. The central claim is that this chain does *not* consist of steps of the same kind. It splits into two qualitatively distinct operations. The first step ($D4 \rightarrow D3$) is not a lossy projection but a *duality decompactification*: the compact mass dimension S_m^1 is unrolled into the non-compact time direction via the fundamental relation $\tilde{T} \cdot m = 1$. This operation preserves the mode content; only the direct visibility of compactness survives, as a discrete spectral signature encoded by the recursion ξ^n . The further steps ($D3 \rightarrow D2 \rightarrow D1 \rightarrow D0$) are genuine geometric projections of spatial directions: strictly lossy, not reversible, and *not* compensated by the recursion. The step $D3 \rightarrow D2$ is identified as a holographic projection. The topological identification $t \sim t + R_m$ is shown to be a physical observable (T_{rev} , CHSH deviation $\sim \xi$). The downward reduction is contrasted with the *upward* operation: the bijective representational translation $T^4 \leftrightarrow$ Hilbert space/matrix (Type III, Doc. 230/206/231), which is *lossless* – precisely because it adds no geometric extra dimensions but expresses the same 4D content in a higher-dimensional language. The work thereby resolves the apparent tension between “projection loses information” and “compactified / decompactified”, and classifies geometric downward reduction against representational upward translation.

.1 Motivation: the tension in the question

FFGFT describes fundamental physics on a compact 4-torus T^4 with the single dimensionless parameter $\xi = 4/30000$ and the central relation $\tilde{T} \cdot m = 1$. The observable world, by contrast, is a three-dimensional space with time evolution, $\mathbb{R}^3 \times \mathbb{R}_t$.

Three observations on this transition appear to conflict:

- (A) **Loss picture.** Every projection $D \rightarrow D-1$ is not injective and therefore loses information. The chain $T^4 \rightarrow \dots \rightarrow T^0$ is not reversible.
- (B) **Compactness picture.** T^4 is fully compactified; the observable spacetime $\mathbb{R}^3 \times \mathbb{R}_t$ is decompactified.
- (C) **Upward picture.** There already exists a *bijective* translation of T^4 into a higher-dimensional space – the Hilbert space (Doc. 230) – which is *lossless*.

These three pictures cannot be naively superimposed: if the first step were pure loss (A), no *decompactified* time direction (B) could arise from it, since forgetting creates no new dimension; and it would remain puzzling how an upward move (C) can be lossless while the downward move (A) destroys information.

The resolution: there is not one but *three* dimension-changing operations of *two kinds*. *Geometric* operations (Type I: mass-time duality; Type II: spatial projection) reduce the observable world out of T^4 ; the *representational* operation (Type III) translates T^4 upward without loss. The geometric-downward lossy versus representational-upward bijective move – this asymmetry is the central result of the work.

.2 Structure of T^4

Decomposition and radius classes

Definition .2.1 (Fundamental torus structure). The fundamental level of FFGFT is the compact 4-torus

$$T^4 = S_x^1 \times S_y^1 \times S_z^1 \times S_m^1,$$

with three *spatial* circles $S_{x,y,z}^1$ and one *mass circle* S_m^1 . All four components are compact.

Remark .2.2 (The fourth direction as substitute time). The fourth torus direction S_m^1 is not a mass dimension *alongside* a separate time, but the *substitute-time direction* itself. Mass and time are two *readings* of one and the same dimension; the fundamental relation $\tilde{T} \cdot m = 1$ is the translation dictionary:

$$\begin{array}{ccc} \underbrace{S_m^1 \text{ (compact)}} & \xleftrightarrow{\tilde{T} \cdot m = 1} & \underbrace{\mathbb{R}_t \text{ (open)}} \\ \text{mass reading, discrete spectrum} & & \text{time reading, continuous flow} \end{array} .$$

There is no time “in addition” to the four directions; time is the decompact reading of the fourth direction (cf. Doc. 185). The two-type structure depends only on the separation of *small* and *large* compactification radii and is independent of the labelling.

The four directions carry two separated radius classes:

$$R_x = R_y = R_z \sim R_H \quad (\text{cosmological horizon, “large”}), \quad (1)$$

$$R_m \sim \xi \cdot \ell_P \quad (\text{Planck scale, “small”}). \quad (2)$$

Remark .2.3 (Status of the two scales). R_H here is the *declared external* (imported) cosmological scale, *not* derived from ξ . Intrinsic is only the *form* – the large/small radius-class split, equivalently the dimensionless ratio R_H/ℓ_P – not the absolute *factor* (Doc. 190, P20). The two-type structure of this document depends only on that class split, not on the numerical value of R_H . Conversely $R_m \sim \xi \ell_P$ sets the *top* of the KK tower ($1/R_m \sim \xi^{-1} m_P$, trans-Planckian); the observed particle masses lie further down the ladder via the recursion ξ^n (Doc. 159, 258/259).

Kaluza–Klein reading

A compact circle of radius R carries a discrete mass tower $m_n \sim n/R$, $n \in \mathbb{Z}$. The two radius classes therefore generate qualitatively different physics:

Circle	Radius	Mode tower
$S_{x,y,z}^1$	$R \sim R_H$ (large)	gap $\sim 1/R_H \approx 0$, quasi-continuous $\Rightarrow$ space
S_m^1	$R \sim \xi \ell_P$ (small)	gap huge, discrete and heavy $\Rightarrow$ mass spectrum

Proposition .2.4 ($\tilde{T}m = 1$ as compactification relation). *In natural units a compactification radius R has the dimension of a length or intrinsic time, $\tilde{T} \sim R$, while the generated mass scale is $m \sim 1/R$. The fundamental relation*

$$\boxed{\tilde{T} \cdot m = 1}$$

is thus structurally the Kaluza–Klein relation for the mass dimension.

.3 Two types of reduction

The *downward* direction (reduction of T^4) proceeds via two geometric operations, developed here. The lossless *upward* operation (Type III, representational translation) follows in Section 8.

Type I: duality decompactification (mass $\rightarrow$ time)

Definition .3.1 (Unrolling the mass dimension). The operation

$$S_m^1 \xrightarrow{\text{covering}} \mathbb{R}_t, \quad p : \mathbb{R}_t \rightarrow S_m^1, \quad t \mapsto e^{2\pi i t/R_m},$$

unrolls the compact mass circle into the non-compact time direction via its universal cover. It is governed by $\tilde{T} \cdot m = 1$.

Theorem .3.2 (Type I preserves mode content). *The unrolling $S_m^1 \rightarrow \mathbb{R}_t$ is an embedding, not a projection. Periodic structures on S_m^1 lift uniquely into $\mathbb{R}_t$; the function space on $\mathbb{R}_t$ is strictly larger. No mode content is forgotten; rather, an additional non-compact direction arises – time.*

Proof: (Sketch) The universal cover $p : \mathbb{R} \rightarrow S^1$ is a local homeomorphism and surjective. An R_m -periodic function on $\mathbb{R}$ corresponds bijectively to a function on S_m^1 (Fourier series, discrete spectrum). The reverse direction (compactification $\mathbb{R} \rightarrow S^1$) is lossy, since non-periodic parts do not descend. Unrolling therefore proceeds in the *embedding*, not the *projecting*, direction.

Corollary .3.3 (Time as embedding price). *The “price” of decompactification is not loss of information but the appearance of the time direction itself (Doc. 185). What does not remain directly visible is only the compactness: the periodicity of S_m^1 appears in the observable not as geometry but indirectly – as the discrete structure of the spectrum.*

Remark .3.4 (What “not totally loss-free” refers to). This is not about absolute loss but about *reading-dependent visibility*. In the mass reading the compactness is manifest – the spectrum is discrete and the identification $t \sim t + R_m$ is expressed directly. In the time reading the same compactness is only *latent*: from continuous time alone the topological identification is not uniquely reconstructible (the winding remains ambiguous). The duality guarantees full translatability of the spectrum; the identification is carried by the time reading only implicitly – through the discreteness of the spectrum. This is exactly where ξ^n acts.

Type II: geometric projection (spatial)

Definition .3.5 (Forgetting a spatial direction). A projection $\pi : \mathbb{R}^k \rightarrow \mathbb{R}^{k-1}$ with $\ker(\pi) = \mathbb{R}$ forgets a spatial direction. It is not injective and therefore not invertible.

Theorem .3.6 (Type II is strictly lossy). *In a spatial projection $\mathbb{R}^k \rightarrow \mathbb{R}^{k-1}$ a full fibre $\mathbb{R}$ collapses. The information about the forgotten direction is not contained in the image and cannot be reconstructed without additional data (boundary conditions, holographic completeness). The recursion ξ^n does not act on this step.*

	Type I (mass $\rightarrow$ time)	Type II (spatial)
Operation	unrolling / covering	forgetting / projection
Direction	embedding	projecting
Mode content	preserved	lost
Reversibility	dual/spectral recoverable	not reversible
Role of ξ^n	spectral signature	ineffective
Residue	compactness only inferred	genuine geometric loss

.4 The formal chain, step by step

Step 1: $D4 \rightarrow D3 + \text{time}$ (Type I)

$$T^4 \longrightarrow \underbrace{T^3}_{\rightarrow \mathbb{R}^3} \times \underbrace{\mathbb{R}_t}_{\text{from } S_m^1}.$$

Step 1 bundles two information-preserving operations of *different* mechanism: (Ia) the *mass circle* unrolls to $\mathbb{R}_t$ via the duality $\tilde{T}m = 1$ per Theorem .3.2 (duality decompactification); (Ib) the three *spatial circles* decompactify through the *limiting process* of large radii ($R_{x,y,z} \rightarrow \infty$ for $r \ll R_H$) to $\mathbb{R}^3$ (limit decompactification). Both are embeddings, not projections; but (Ib) is *not* a duality step, it is a limit – see the clarification in Section .12, point 3.

Invariant: $\tilde{T} \cdot m = 1$ (Prop. .2.4); in the spatial limit, local flatness. *Loss:* none in mode content; only the visibility of compactness (Rem. .3.4).

Step 2: $D3 \rightarrow D2$ (Type II, holographic)

$$\pi_2 : \mathbb{R}^3 \longrightarrow \mathbb{R}^2, \quad \ker(\pi_2) = \mathbb{R} \text{ (depth direction).}$$

Invariant: area A . *Holography:* step 2 is mathematically the holographic principle; in the horizon case the Bekenstein–Hawking bound $S \leq A/(4\ell_P^2)$ (cf. Doc. 254/268). *Loss:* genuine depth direction.

Step 3: $D2 \rightarrow D1$ (Type II)

$$\pi_3 : \mathbb{R}^2 \longrightarrow \mathbb{R}, \quad \ker(\pi_3) = \mathbb{R}.$$

Invariant: length L . *Loss:* second spatial direction; no area measure remains.

Step 4: $D1 \rightarrow D0$ (Type II, total loss)

$$\pi_4 : \mathbb{R} \longrightarrow \{\text{pt}\}.$$

Invariant: only a discrete scalar measure ($n \in \mathbb{Z}$). *Loss:* the complete geometric structure.

.5 Reversibility, precise per type

Theorem .5.1 (Reversibility of the full chain). *Let $\Pi = \pi_4 \circ \pi_3 \circ \pi_2$ be the spatial subchain $\mathbb{R}^3 \rightarrow \{\text{pt}\}$ and D the Type I operation $S_m^1 \rightarrow \mathbb{R}_t$. Then:*

- (i) *D is, as a duality (Fourier/KK), invertible in mode content; the winding remains ambiguous. D is “almost reversible”.*
- (ii) *Π is not invertible: there is no continuous map $\rho : \{\text{pt}\} \rightarrow \mathbb{R}^3$ with $\rho \circ \Pi = \text{id}$.*

Proof: (Sketch) (i) follows from Theorem .3.2: unrolling is embedding, the discrete spectrum encodes the full periodic content; only the choice of lift (winding) is non-unique. (ii) follows from Theorem .3.6: each π_k has $\ker = \mathbb{R} \neq \{0\}$. A left inverse would have to reconstruct the collapsed fibre structure; this information is not present in the image $\{\text{pt}\}$.

Corollary .5.2 (Resolution of the initial tension). *Pictures (A) and (B) are both correct but concern different operations. The non-reversibility (A) holds for the spatial projections (Type II). The decompactification to the time direction (B) is the embedding Type I step – it creates a dimension instead of forgetting one. There is no contradiction.*

.6 Role of the recursion ξ^n

The recursion $R : \xi^n \mapsto \xi^{n+1}$ is *not* a universal compensation for projection losses. Its function is narrowly bounded:

- **Type I:** ξ^n generates the discrete mass spectrum (particle ladder). This spectrum is the compact mode content of S_m^1 , dually expressed as time-frequency structure. ξ^n “fills in” nothing; it is the spectral signature through which the hidden compactness remains recognisable (Rem. .3.4).
- **Type II:** ξ^n has no effect. Spatial projection losses are genuine and untouched by the recursion.

Remark .6.1 (Correction of a tempting misrepresentation). A presentation that assigns the recursion a “compensation strength” decreasing monotonically across all four steps (full/partial/weak/none) is misleading. It presupposes a uniform lossy ladder that does not exist. The recursion belongs exclusively to the Type I step.

.7 Overview

Step	Operation	Type	Invariant	ξ^n
$D4 \rightarrow D3+t$	$S_m^1 \rightarrow \mathbb{R}_t$ (unroll)	I	$\tilde{T}m = 1$	spectral signature
$D3 \rightarrow D2$	forget space	II	area A	–
$D2 \rightarrow D1$	forget space	II	length L	–
$D1 \rightarrow D0$	forget space	II	$n \in \mathbb{Z}$	–

Table 1: The chain $T^4 \rightarrow T^0$ separated by type.

$$\begin{array}{ccc}
 \underbrace{T^4 \longrightarrow \mathbb{R}^3 \times \mathbb{R}_t}_{\text{Type I: embedding, almost reversible}} & \longrightarrow & \underbrace{\mathbb{R}^2 \longrightarrow \mathbb{R} \longrightarrow \{\text{pt}\}}_{\text{Type II: projecting, not reversible}}
 \end{array}$$

.8 The upward direction: representational translation (Type III)

The chain so far describes only the *downward* direction (reduction). There is a counter-direction – but it is *not* geometric.

Definition .8.1 (Type III: representational translation). The operation

$$T^4 \xrightleftharpoons{\text{bijective}} \mathcal{H}$$

translates the FFGFT mode formalism on T^4 into the (infinite-dimensional) Hilbert space $\mathcal{H}$, or the equivalent matrix algebra (Doc. 230, 206, 231). It is not an approximation but a *bijective structural statement*: every QM computation is expressible in FFGFT language and vice versa.

The dimensional asymmetry

Downward (geometric) and upward (representational) are different kinds of operation with *opposite* information properties:

- **Downward (Type II, geometric):** lossy, not reversible – a fibre collapses (Theorem .3.6).

- **Upward (Type III, representational):** bijective, lossless – it adds *no* physical dimensions but expresses the same 4D content in a higher-dimensional *language* in which it fits without reduction.

The losslessness follows *precisely from* the representational character – not from any gain of genuine extra dimensions.

Remark .8.2 (No geometrically higher dimensions). A geometric extension $T^4 \rightarrow T^5 \rightarrow \dots$ is excluded in FFGFT (Doc. 171): complexity arises from *fractal depth* ($D_f = 3 - \xi$, self-similarity), not from *more* dimensions. This is the deliberate contrast with string theory (10/11D, Calabi-Yau, landscape problem). “Higher” therefore always means Type III (representational) in FFGFT, never geometric.

FFGFT thus has three dimension-changing operations:

Type	Operation	Character	Information
I	mass $\leftrightarrow$ time ($\tilde{T}m = 1$)	geometric	almost reversible
II	spatial projection (down)	geometric	lossy
III	$T^4 \leftrightarrow \mathcal{H}$ /matrix (up)	representational	bijective

The bijectivity of Type III is the mirror image of the non-reversibility of Type II (Theorem .5.1): geometrically one loses on reducing, representationally one loses nothing on translating.

.9 Why exactly four circles? Minimal sufficiency, units, and visualization

The chain so far (Type I/II downward, Type III upward) presupposes the four-circle structure $T^4 = S_{x,y,z}^1 \times S_m^1$. This section answers *why* there are exactly four – deliberately *not* as a derivation of the number, but as *minimal sufficiency*.

3 + 1 is input, not derivation

No theory derives the dimension count of spacetime: the Standard Model and GR assume 3 + 1; string theory fails on uniqueness (landscape problem). FFGFT, too, *does not derive the 4 forward*; it takes 3 + 1 as *empirical input*. The question is therefore not “why is the world four-dimensional?” but “*given* 3 + 1 – which structure is the minimal one that carries everything?”.

Minimal sufficiency: four instead of five

The duality saves one dimension

Naively one would need *five* building blocks: three for space, one for a mass spectrum (particles), one for time evolution. The fundamental relation $\tilde{T} \cdot m = 1$, however, identifies the *mass reading* and the *time reading* of *one* dimension (Type I, Theorem .3.2): the fourth circle S_m^1 carries the mass spectrum as its KK tower *and*, decompactified, time. The duality lets *one* circle do the work of two – so the naive 5 collapses to **4**.

- **Fewer** (only T^3): no mass spectrum, no time.
- **Four** (T^3 + one dual circle): space, masses *and* time from *one* object, without redundancy.

- **More** (a second such circle): a second independent mass tower or a second time – superfluous, without correlate.

“Optimal” therefore means *minimal-sufficient*: the smallest dimension count in which space, the full mass spectrum, and time jointly emerge from one geometric object.

The four circles as units bookkeeping

The four-circle structure *is* at the same time the exponent bookkeeping of the SI conversion factors – each circle type carries its bridge:

$$\underbrace{S_{x,y,z}^1}_{\text{space}} \xrightarrow{c} \text{time}, \quad \underbrace{S_m^1}_{\text{mass}} \xrightarrow{\hbar (\tilde{T}_m=1)} \text{time}.$$

c is the space-time bridge of the three spatial circles, $\hbar$ the mass-time bridge of the mass circle. This is why the SI conversions are *indispensable* once one wants to determine the exponent structure correctly:

Remark .9.1 (Why natural units hide the powers). Natural units ($c = \hbar = 1$) collapse L , T and M to *one* axis. This removes precisely the distinction that carries the exponent structure – the independent *power of c* resp. *of time*. c and $\hbar$ may be set to 1 only *after* the exponents have been read off; before that the information is destroyed. The factors are the scaffolding that keeps the powers visible – and easy to handle precisely because their whole role is that scaffolding without a free parameter. G , too, is *derived* – reconstructed from ℓ_P , c , $\hbar$, for which it needs only the SI relations (Docs. 012/056) – and carries *no* free factor. The only free factor sits not in a conversion or natural constant but in the *cosmological absolute scale* R_H (exponent 41/4; Doc. 190, R20).

The same structure as Hilbert visualization

The four-circle geometry is also the *faithful* visualization of the infinite-dimensional Hilbert space (Type III, Def. .8.1) – two readings of *the same* bijection:

- **generative/ontological**: $T^4 \rightarrow \mathcal{H}$ – the geometry *generates* the algebra (Doc. 206), not the reverse;
- **visualizing/epistemic**: the abstract $\mathcal{H}$ becomes intuitable at all only through the 4D geometry.

“Higher-dimensional” (the ∞ -dimensional $\mathcal{H}$) does *not* mean “more fundamental”: the generative direction is $T^4 \rightarrow \mathcal{H}$. Precisely *because* the 4D structure is generative (not a mere picture), the visualization is *exact* – bijective, not a cartoon.

Summary of the dimension line

What “optimal” means

T^4 resp. $T^3 + \text{time}$ is the *minimal-sufficient*, *dual-economical* and *Hilbert-faithful* configuration – *given* $3 + 1$. The strength lies in the economy (one dimension saved by $\tilde{T}_m = 1$) and the faithfulness (lossless Hilbert representation), *not* in a uniqueness of the number. The bare 4 is input, not a forward result (Doc. 190, R34) – the same signature as for the ξ -magnitude (R33) and the cosmological scale (R20): *structure forward, number input*.

.10 In the compact view, time is just another circle: the asymmetry is the measurement

In the compact representation $T^4 = S^1 \times S^1 \times S^1 \times S^1$ all four circles are identical: dimensionless cyclic coordinates, no signature, no space/time distinction. Time is written *exactly* as the three spatial circles; at the compact level the Lorentzian asymmetry does not exist. This is why, in FFGFT, the time coordinate is simple to write – it is one circle like the others. The complication enters only when one leaves the compact picture.

The asymmetry (one time, signature (3, 1)) appears only under decompactification (Type I, Def. .3.1): exactly one circle – the mass circle, $\tilde{T} \cdot m = 1$ – unrolls through its universal cover into non-compact time, while the other three stay compact (discrete spectrum) or project (space, Type II).

Which circle becomes time is therefore not an intrinsic-geometric fact to be derived. The geometry is symmetric, and nothing in it should single one direction out. Time is *relational*: the selection is the measurement itself. The decompactification *is* the measurement – the read-out/projection of this document and of Doc. 284 – and committing one circle to non-compact, experienced time is exactly the information-discarding step a measurement performs. The embedded observer, reading out from the inside (Doc. 248, self-referentiality), experiences one direction as duration; that experience constitutes the time circle. The symmetry-breaking is the measurement, not the geometry.

[The time circle is selected by the measurement] In the compact view the four circles of T^4 are symmetric. Time is not chosen by the geometry but is relational: the symmetry-breaking *is* the measurement (the decompactification of Type I). Three statements then close the bookkeeping with no gap:

- **That there is one time** (not two) is the signature/predictivity constraint – one timelike direction gives hyperbolic field equations with a well-posed Cauchy problem; two or more give ultrahyperbolic equations with no well-posed initial-value problem.
- **Which circle is time** is fixed by the measurement, i.e. by the experienced situation of the embedded observer (Doc. 248).
- **That the split is $3+1$** is the structure of being an embedded observer – the empirical input of Doc. 190 (P34), read here as the observer's situation rather than a free parameter.

The contrast with a non-compactified treatment of time is then sharp. A framework that writes time as a complicated structure built on already-physical time t (for instance a spiral-time operator $\psi(t) = t + i\varphi(t) + j\chi(t)$, Doc. 283/284) has *reified* the result of the measurement into its kinematics, instead of recognising experienced time as the measurement-induced decompactification of a symmetric circle. In the compact view that complication is unnecessary: time is one more circle, and only the read-out turns it into the non-periodic axis we experience.

.11 Physical testability of the identification $t \sim t + R_m$

The topological identification $t \sim t + R_m$ is not a mere accompanying structure of the duality but a physical observable. In the time reading its period is *exactly* the fundamental revolution time

$$T_{\text{rev}} = \frac{R_m}{c} = \frac{\xi \ell_P}{c} \approx 7.2 \times 10^{-48} \text{ s}, \quad \Delta\phi = 2\pi\xi \approx 8.4 \times 10^{-4} \text{ rad/revolution},$$

so that $t \sim t + R_m$ is identical with $t \sim t + cT_{\text{rev}}$ (Doc. 183/185). The compactness of the fourth direction *is* the revival time.

Two testable readings

- **Mass reading (mature, quantitative).** The discrete mode spectrum generates the particle masses as torus overtones with step $1/D$; from this follow Koide, $g-2$ and the three generations (Doc. 159, 258/259). Here the compactness is numerically realised.
- **Time reading (falsifiable, currently below resolution).** The periodicity produces a CHSH deviation from the Tsirelson bound of order ξ . For 73 qubits FFGFT predicts $\Delta \approx 5.4 \times 10^{-4}$ (resolvable at $\sigma \sim 10^{-4}$). The hardware test IBM Heron r2 (May 2026) confirms the Bell violation ($S = 2.74$) and the executability of T0 circuits; the ξ effect, however, lies below current NISQ noise and is not yet resolvable (Doc. 147).

Scaling: linear versus logarithmic

Regime dependence of phase accumulation

A naive *coherent linear accumulation* $r \cdot \Delta\phi$ over r revolutions would yield physically absurd phase drifts (of order 10^{44} rad/s) and contradict working atomic clocks and deep quantum circuits. The lab-testable prediction therefore scales *logarithmically*,

$$\delta\phi \sim \frac{\xi \ln(r)}{D_f} \approx 2 \times 10^{-4} \quad (D_f = 3 - \xi),$$

not linearly. The linear form $r \cdot \Delta\phi$ of Doc. 185 is the *worst-case bound of the RSA argument* (coherence over $r \approx N$ cycles), not the lab prediction. This regime separation explains why deep coherent circuits can run while the RSA factorisation of large keys remains geometrically protected.

Conclusion. $t \sim t + R_m$ is carried in the FFGFT corpus as an observable (T_{rev} , CHSH deviation $\sim \xi$): quantitatively confirmed in the mass reading, falsifiable in the time reading but currently just below experimental resolution. It is not an unfalsifiable accompanying structure.

.12 Classification and open questions

Connections. Doc. 185 (Time as the Embedding Price) provides the ontological basis of the Type I step; Doc. 268 (holographic projection) and Doc. 254 (area-bound theorem) the framework for step 2; Doc. 206 (Triangle–Matrix reduction) confirms that the 4D level is generative; Doc. 147/183 provide the time-side testability. The upward operation (Type III) is developed in Doc. 230 (bijective Hilbert-space translation), Doc. 231 (necessary extensions) and Doc. 206 (matrix representation); Doc. 171 justifies the exclusion of geometrically higher dimensions.

Open points.

1. **Structural assumption – confirmed.** The decomposition $T^4 = S_{x,y,z}^1 \times S_m^1$ with S_m^1 as substitute-time direction and $\tilde{T} \cdot m = 1$ as translation is fixed (Rem. .2.2).

2. **Linear/logarithmic scaling.** The linear worst-case bound $r \cdot \Delta\phi$ (Doc. 185, RSA regime) and the logarithmic lab prediction $\xi \ln(r)/D_f$ (Doc. 147) should be marked explicitly as regime-dependent to secure consistency of presentation.
3. **Spatial limit – clarified.** The decompactification $S_{x,y,z}^1 \rightarrow \mathbb{R}^3$ is a *limiting process* ($R_{x,y,z} \rightarrow \infty$), neither the Type I duality nor a Type II projection, but a distinct *limit-type* information-preserving operation (Ib). It is therefore carried in Step 1 explicitly apart from the mass-circle duality (Ia); both share the embedding character, not the mechanism.
4. **Holographic completeness.** To what extent can the Type II loss at $D3 \rightarrow D2$ be reconstructed exactly (not just partially) from boundary data? Connection to Doc. 254/268.